



Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Exercise Sheet 5

Training Exercises

Exercise T5.1. *Modelling Tricks - OR*

Let $x, y \geq 0$ and $x, y \in \mathbb{Z}$. How can we formulate conditions such that either $x + y \leq 3$ or $2x + 5y \leq 12$ (or both) always hold? How to model it so that exactly one of these conditions holds (XOR)?

Exercise T5.2. *Caffeine*

Maxime is often very tired at work. She describes her tiredness by a *tiredness level*. This is a natural number from the set $\{0, \dots, 20\}$, where 20 means very tired, and 0 means wide awake. To counteract her tiredness, she can buy various caffeinated drinks:

Drink	Decrease in tiredness	Price (euro)
Coffee	11	3
Black Tea	7	2
Green Tee	3	1

Maxime works 8 hours each Monday through Friday. In each working hour, she can choose to consume none or exactly one of the drinks. If she chooses to drink none, her tiredness level increases by 2 during that hour. If she chooses to drink one of the drinks, her tiredness level decreases by the corresponding value in the *Decrease in tiredness* column. The following additional conditions must be met:

- The tiredness level on Mondays in the first hour of work is 10.
- For any other workday, if any of the drinks were consumed during the 6th, 7th, or 8th hour of work the day before, Maxime starts the day with a fatigue level of 13, otherwise 10.
- Maxime can't spend more than 30 euros a week on drinks.
- Maxime can't drink a beverage that would make her tiredness level drop below 0.
- She must drink one of the three drinks once her fatigue level is greater than 18.
- Maxime doesn't want to drink coffee at least one day a week.

Maxime wants to minimize the sum over all tiredness levels of all hours worked in a week. Formulate this problem as IP without solving it.

Exercise T5.3. Exam Preparation

The teaching assistants Georgina and Maxime are planning the exam for the lecture “Molecular Biology for Information Systems”. They have created a collection of 20 potential *tasks* $a \in A := \{1, \dots, 20\}$ and assigned to each of these tasks a *complexity* $k_a \in \{1, \dots, 10\}$ as well as an *estimated completion time in minutes* $d_a \in \mathbb{N}$. Furthermore, the tasks can be classified into two *types*: calculus tasks and multiple choice tasks. The parameter $t_a \in \{0, 1\}$ describes the type of task $a \in A$, where $t_a = 0$ means that a is a calculus task, while $t_a = 1$ means that a is a multiple choice task. The exam may consist of *at most* 12 tasks.

Georgina and Maxime would like to leave the design of what they consider the most optimal exam to Gurobi. They therefore introduce binary variables $x_{as} \in \{0, 1\}$ for $a \in A$ and $s \in S := \{1, \dots, 12\}$, where $x_{as} = 1$ indicates that task a is on the s -th page of the exam. The exam must satisfy several constraints. Formulate the following constraints as linear (in)equalities. You *may* introduce additional variables. If you do, state the intuitive meaning of the variables.

- a) There may be no more than one task on each page $s \in S$ of the exam.
- b) Each task may be in the exam no more than once.
- c) The total complexity of the exam is the sum of the complexities of the tasks it contains and must be at least 40.
- d) The estimated time required to complete the exam must not exceed 115 minutes.
- e) No three tasks of the same type may follow each other consecutively.
- f) The exam should not have a blank page: If there is no task on the s -th page of the exam, then there is no task on all subsequent pages $s' > s$.
- g) If the exam contains an easy task with complexity $k_a \leq 2$, then it must also contain at least one difficult task with complexity $k_a \geq 8$.
- h) Georgina and Maxime have two objectives for the exam: (1) the exam should have a complexity of 60 and (2) the exam should have a completion time of 95 minutes. The complexity objective is twice as important to them as the completion time objective. Formulate an objective function that aggregates both objectives. Introduce additional variables and constraints if necessary.

Self-study Exercises

Exercise S5.1. *Logic*

Using integer linear programs, it is easy to impose logical conditions on the variables. In the following, $A \in \{0, 1\}$ and $B \in \{0, 1\}$ denote binary variables, where as usual 1 corresponds to *true* and 0 to *false*.

- a) The binary variable $Z_{\neg A} \in \{0, 1\}$ is said to express the negation of A , so $Z_{\neg A} = 1 \Leftrightarrow A = 0$. Express this using linear (in-)equalities.
- b) The binary variable $Z_{A \vee B} \in \{0, 1\}$ is said to express the disjunction of A and B , i.e. $Z_{A \vee B} = 1 \Leftrightarrow A \vee B = 1$. Express this using linear (in-)equalities.
- c) Do the same for the conjunction $A \wedge B$ and the implication $A \rightarrow B$.

Also, visualize the feasible set in each case.

Now let $P \in [0, 100]$ be a real-valued variable.

- d) For the binary variable Z let $P < 10 \Rightarrow Z = 1$ and $P > 10 \Rightarrow Z = 0$ hold. Again, express this using linear (in-)equalities.
- e) Is it possible to define a variable Z with $Z = 1 \Leftrightarrow P \leq 10$?
Hint: What would the corresponding polytope have to look like?

Exercise S5.2. *Party Planning*

To celebrate your successful final exam in OR, you spontaneously decide to throw a party. To do this, you want to buy drinks from the following selection at the nearby supermarket. Note that you can only buy whole bottles at the supermarket!

Beverage	Price per bottle	Volume per bottle
Sparkling water	0,5 €	1 l
Orange juice	0,7 €	1 l
Vodka	7 €	0.7 l
Beer	0,8 €	0.5 l

You decide to meet the following constraints:

- You want to buy at least 10 liters of non-alcoholic drinks (sparkling water or orange juice).
- You want to buy at least 10 bottles of beer and one bottle of vodka.
- You want to buy at least 40 bottles of beer or at least 5 bottles of vodka. (OR)
- When you buy vodka, in addition to the orange juice purchased for non-alcoholic use, you want to buy another 2 bottles of orange juice per bottle of vodka.
- For weight reasons, you want to buy at most 30 liters of beverages. However, the supermarket offers you free home delivery for purchases of 50 € or more. In this case, you can buy as much as you want without having to worry about the weight.

Formulate a mixed-integer linear program to buy drinks for your party as cheap as possible while satisfying this set of constraints.

Exercise S5.3. *Dutch Petroleum*

Dutch Petroleum (DP) produces two different types of fuel and requires two different types of crude oil for this. The following quality requirements apply.

- Fuel 1: Minimum crude oil type 1: 50%
- Fuel 2: Minimum crude oil type 1: 60%

DP currently holds 10 000l of crude oil 1 and 20 000l of crude oil 2. DP is able to purchase a maximum of 30 000l of crude oil type 1 at the following prices, whereby DP receives a volume discount for large purchases: The price is 1.8€/l if less than 10 000l is purchased and 1.6€/l if more than 10 000l is purchased.

Fuel 1 can be sold for 1.4€/l, fuel 2 even for 1.5€/l.

Formulate an integer linear optimization model to solve this problem, with the aim of optimizing the profit (i.e., revenue - costs).